



Time Series Analysis

Seminar 10: State space models, Kalman filter and Markov switching



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Acronyms used in this material

- ▣ **AI:** Artificial Intelligence
- ▣ **AR:** AutoRegressive (model)
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BIC:** Bayesian Information Criterion
- ▣ **CBO:** Congressional Budget Office
- ▣ **EODHD:** EOD Historical Data (market-data provider)
- ▣ **FRED:** Federal Reserve Economic Data
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP:** Gross Domestic Product
- ▣ **HICP:** Harmonised Index of Consumer Prices
- ▣ **HP:** Hodrick–Prescott (filter)
- ▣ **NBER:** National Bureau of Economic Research
- ▣ **OLS:** Ordinary Least Squares
- ▣ **SD:** Standard Deviation
- ▣ **SE:** Standard Error
- ▣ **SES:** Simple Exponential Smoothing
- ▣ **UC:** Unobserved Components (model)
- ▣ **US:** United States
- ▣ **VAT:** Value Added Tax

Today's question and route

- **Question:** how do we estimate a quantity we never observe (a level, a trend, a regime) from noisy data, one period at a time?
 - ▶ this seminar comes **before** Lecture 10: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: the Kalman filter by hand; exponential smoothing as a steady state; state space forms; durations; the Hamilton filter
 - ▶ Part B: Romanian inflation; the US output gap; US industrial production and recessions; BET volatility regimes
 - ▶ Part C: regimes of Romanian inflation since 1996; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	the Kalman filter of a local level model for three periods; a missing value	Solved, Proposed	A1
A3, A4	steady state and exponential smoothing; state space forms and forecasts	Solved, Proposed	A3
A5, A6	durations of regimes; one step of the Hamilton filter	Solved, Proposed	A5
B1, B2	underlying Romanian inflation; the US output gap	Solved, Proposed	B1
B3, B4	recessions in US industrial production; BET volatility regimes	Solved, Proposed	B3
C1, C2	Romanian inflation regimes; what is wrong in an AI answer?	Proposed	B3, A1–A6

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	1996–2026
real GDP and potential GDP, US	FRED (GDPC1, GDPPOT)	quarterly	1947–2026
industrial production, US	FRED (INDPRO)	monthly	1960–2019
NBER recession indicator	FRED (USREC)	monthly	1960–2019
BET, S&P 500	EODHD	weekly, from daily closes	2000–2026

- Monthly inflation: $100\Delta \ln P_t$, seasonally adjusted (SA) by removing the monthly means ([Chapter 4](#)); CBO output gap: $100(\text{GDP}_t/\text{GDPPOT}_t - 1)$
- In the notebook: `kalman_by_hand`, `local_level_ml`, `uc_fit`, `hamilton_filter`, `MarkovRegression`; no account or key is needed

Prerequisites for Today (1/5): the state space form

- **State space form:** a hidden state α_t evolves over time and is observed with noise
 - ▶ measurement: $y_t = Z\alpha_t + \varepsilon_t$, $\varepsilon_t \sim N(0, H)$; transition: $\alpha_{t+1} = T\alpha_t + R\eta_t$, $\eta_t \sim N(0, Q)$
 - ▶ Z : links the state to the observation; T : moves the state one period ahead; R : maps the shocks into the state; H , Q : the variances of the two shocks
 - ▶ forecasts iterate the transition: $\hat{\alpha}_{T+h} = T^h \hat{\alpha}_T$, with $\hat{\alpha}_T$ the estimate of the last state

Prerequisites for Today (2/5): four models in state space form

- ▣ **Local level:** $y_t = \mu_t + \varepsilon_t$, $\mu_{t+1} = \mu_t + \eta_t$; μ_t : the level (a random walk)
 - ▶ signal-to-noise ratio $q = \sigma_\eta^2 / \sigma_\varepsilon^2$: the variance of the level shocks relative to the measurement noise
- ▣ **Local linear trend:** $\mu_{t+1} = \mu_t + \beta_t + \eta_t$, $\beta_{t+1} = \beta_t + \zeta_t$; state $(\mu_t, \beta_t)'$
 - ▶ β_t : the slope (growth per period); ζ_t : the shock to the slope
- ▣ **AR(2):** $\alpha_t = (y_t, y_{t-1})'$, $T = \begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix}$, $Z = (1, 0)$, $H = 0$
- ▣ **Simple exponential smoothing (SES, [Chapter 0](#)):** $\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$, forecast $\hat{y}_{t+1} = \ell_t$
 - ▶ ℓ_t : the smoothed level; $\alpha \in (0, 1)$: the weight of the newest observation

Prerequisites for Today (3/5): the Kalman filter of the local level model

- Start with a prediction a_t of μ_t and its variance P_t ; when y_t arrives [Kalman \(1960\)](#):
 - ▶ $v_t = y_t - a_t$ (surprise), $F_t = P_t + \sigma_\varepsilon^2$ (its variance), $K_t = P_t/F_t$ (**Kalman gain**, between 0 and 1)
 - ▶ update: $a_{t|t} = a_t + K_t v_t$, $P_{t|t} = P_t(1 - K_t)$
 - ▶ prediction: $a_{t+1} = a_{t|t}$, $P_{t+1} = P_{t|t} + \sigma_\eta^2$
- **Missing** y_t : no update, $a_{t|t} = a_t$, $P_{t|t} = P_t$; after the last observation y_n , forecasts of every horizon equal a_{n+1}
 - ▶ **steady state**: $\bar{P} = \sigma_\varepsilon^2(q + \sqrt{q^2 + 4q})/2$, $\bar{K} = \bar{P}/(\bar{P} + \sigma_\varepsilon^2)$; then $a_{t+1} = \bar{K}y_t + (1 - \bar{K})a_t$ is SES with $\alpha = \bar{K}$ [Muth \(1960\)](#)
 - ▶ inverse map: $q = \alpha^2/(1 - \alpha)$

Prerequisites for Today (4/5): likelihood, smoothing, trend and cycle

- **Likelihood** (prediction-error decomposition): $\log L = -\frac{1}{2} \sum_t (\log 2\pi + \log F_t + v_t^2/F_t)$, maximised over the variances
 - ▶ large surprises v_t relative to their variance F_t lower the likelihood
 - ▶ **filtered** $a_{t|t}$ uses $y_1, \dots, y_t$ (real time); **smoothed** $\hat{\mu}_t$ uses the whole sample (history) [Durbin and Koopman \(2012\)](#)
- **Output gap**: the cycle ψ_t in $100 \log \text{GDP}_t = \mu_t + \psi_t$ (trend plus cycle)
 - ▶ **UC model** [Harvey \(1989\)](#): smooth trend ($\Delta^2 \mu_{t+1} = \zeta_t$, Δ^2 : the second difference) plus an AR(2) cycle, estimated by the Kalman filter
 - ▶ **HP filter** [Hodrick and Prescott \(1997\)](#): the trend τ_t minimises $\sum (y_t - \tau_t)^2 + 1600 \sum (\Delta^2 \tau_t)^2$, a two-sided smoother
 - ▶ **Hamilton filter** [Hamilton \(2018\)](#): residual of OLS of y_{t+8} on $1, y_t, \dots, y_{t-3}$ ($y_t = 100 \log \text{GDP}_t$)

Prerequisites for Today (5/5): Markov switching

- **Markov switching Hamilton (1989):** $y_t = \mu_{S_t} + \varepsilon_t$, $\varepsilon_t \sim N(0, \sigma_{S_t}^2)$, regime $S_t \in \{1, 2\}$ follows a Markov chain
 - ▶ μ_j, σ_j : mean and standard deviation in regime j ; $p_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$, collected in the transition matrix $\mathbf{P}$
 - ▶ **expected duration** of regime i : $1/(1 - p_{ii})$; **ergodic probability** $\pi_1 = (1 - p_{22})/(2 - p_{11} - p_{22})$; h -step transitions: $\mathbf{P}^h$
- **Hamilton filter:** predict $\Pr(S_t = 1 \mid Y_{t-1}) = p_{11}\xi_{t-1} + (1 - p_{22})(1 - \xi_{t-1})$, with $\xi_{t-1} = \Pr(S_{t-1} = 1 \mid Y_{t-1})$
 - ▶ Y_{t-1} : the data up to $t - 1$; update with the Normal densities $f_j(y_t) = \varphi((y_t - \mu_j)/\sigma_j)/\sigma_j$ (φ : standard Normal density):
 - ▶ $\xi_t = \frac{\Pr(S_t = 1 \mid Y_{t-1})f_1(y_t)}{\Pr(S_t = 1 \mid Y_{t-1})f_1(y_t) + \Pr(S_t = 2 \mid Y_{t-1})f_2(y_t)}$; **filtered** probabilities for real time, **smoothed** ones for history

A1: the Kalman filter for three periods [Solved]

Task

- Local level with $\sigma_\varepsilon^2 = 1$, $\sigma_\eta^2 = 1$; prior $a_1 = 0$, $P_1 = 1$; data $y = (1, 3, 2)$.
- 1. For $t = 1, 2, 3$ compute F_t , K_t , v_t , $a_{t|t}$, $P_{t|t}$ and the next prediction.
- 2. Give the forecast of y_4 and its variance.
- 3. Compute the steady-state gain $\bar{K}$ for $q = 1$.
- Report: a table of the three steps, one forecast, $\bar{K}$.

Solution

- 1. $t = 1$: $F = 2$, $K = 0.5$, $v = 1$, $a_{1|1} = 0.5$, $P_{1|1} = 0.5$; $a_2 = 0.5$, $P_2 = 1.5$
- $t = 2$: $F = 2.5$, $K = 0.6$, $v = 2.5$, $a_{2|2} = 0.5 + 0.6 \cdot 2.5 = 2$, $P_{2|2} = 0.6$; $a_3 = 2$, $P_3 = 1.6$
- $t = 3$: $F = 2.6$, $K = 0.615$, $v = 0$, $a_{3|3} = 2$, $P_{3|3} = 0.62$
- 2. $\hat{y}_4 = a_4 = 2.00$; $P_4 = 1.62$; $\text{Var} = P_4 + \sigma_\varepsilon^2 = 1.62 + 1 = 2.62$
- 3. $\bar{P} = (1 + \sqrt{5})/2 = 1.618$ (the golden ratio), $\bar{K} = 0.618$; K_t is already 0.615 at $t = 3$



A2: a missing observation [Proposed]

Solution discussed in class

Task

- Local level with $\sigma_\varepsilon^2 = 2$, $\sigma_\eta^2 = 0.5$; $a_1 = 5$, $P_1 = 2$; data $y = (6, \text{missing}, 4)$. Model: A1.
- 1. Run the filter for $t = 1, 2, 3$, skipping the update at $t = 2$.
- 2. Explain why K_3 equals K_1 .
- 3. Compute $\bar{K}$ for this q and the equivalent SES weight.
- Report: the three steps, one sentence, $\bar{K}$.

A3: exponential smoothing is a steady-state Kalman filter [Solved]

Task

- A local level model and simple exponential smoothing (SES).
- 1. Show that the steady-state update $a_{t+1} = a_t + \bar{K}(y_t - a_t)$ is SES.
- 2. Compute $\bar{K}$ for $q = 1$ and $q = 0.25$.
- 3. An SES model has $\alpha = 0.3$. Find the q of the local level model behind it.
- Report: one line of algebra, two gains, one q .

Solution

- 1. $a_{t+1} = \bar{K}y_t + (1 - \bar{K})a_t$: the SES recursion with $\alpha = \bar{K}$ and $\ell_t = a_{t+1}$
- 2. $q = 1$: $\bar{P}/\sigma_\epsilon^2 = (1 + \sqrt{5})/2 = 1.618$,
 $\bar{K} = 1.618/(1.618 + 1) = 0.618$; $q = 0.25$: $\bar{P}/\sigma_\epsilon^2 = 0.640$,
 $\bar{K} = 0.390$
- 3. From $\bar{P} = \bar{P}(1 - \bar{K}) + \sigma_\eta^2$: $\sigma_\eta^2 = \bar{K}\bar{P}$ and
 $\bar{P} = \alpha\sigma_\epsilon^2/(1 - \alpha)$, so $q = \alpha^2/(1 - \alpha) = 0.09/0.7 = 0.129$
- a small α means a level that moves little relative to the noise

A4: state space forms and forecasts [Proposed]

Solution discussed in class

Task

- ▣ An AR(2) with $\phi_1 = 0.5$, $\phi_2 = 0.3$, $\sigma^2 = 1$; a local linear trend. Model: A1 (the prediction step).
- ▣ 1. Write Z , T , R , Q and H for the AR(2).
- ▣ 2. With the last state $(y_T, y_{T-1}) = (2, 1)$, compute the forecasts for $h = 1, 2, 3$ by $\hat{\alpha}_{T+h} = T\hat{\alpha}_{T+h-1}$.
- ▣ 3. Write Z and T of the local linear trend and forecast three steps from $\mu_T = 100$, $\beta_T = 2$.
- ▣ Report: two sets of matrices and six forecasts.

A5: durations of recessions and expansions [Solved]

Task

- ▣ A two-regime model of GDP growth: regime 1 (recession) with $p_{11} = 0.75$, regime 2 (expansion) with $p_{22} = 0.95$.
- ▣ 1. Write the transition matrix and compute the expected durations.
- ▣ 2. Compute the ergodic probabilities.
- ▣ 3. Starting in a recession, compute the probability of a recession two quarters later.
- ▣ Report: two durations, two probabilities, one two-step probability.

Solution

- ▣ 1. $\mathbf{P} = \begin{pmatrix} 0.75 & 0.25 \\ 0.05 & 0.95 \end{pmatrix}$; durations $1/0.25 = 4$ and $1/0.05 = 20$ quarters
- ▣ 2. $\pi_1 = 0.05/(0.25 + 0.05) = 0.167$, $\pi_2 = 0.833$: one quarter in six is a recession quarter
- ▣ 3. $p_{11}^{(2)} = 0.75^2 + 0.25 \cdot 0.05 = 0.575$; the chance of an expansion is 0.425
- ▣ the geometric duration: many recessions are short, a few are long; the mean is 4 quarters

A6: one step of the Hamilton filter [Proposed]

Solution discussed in class

Task

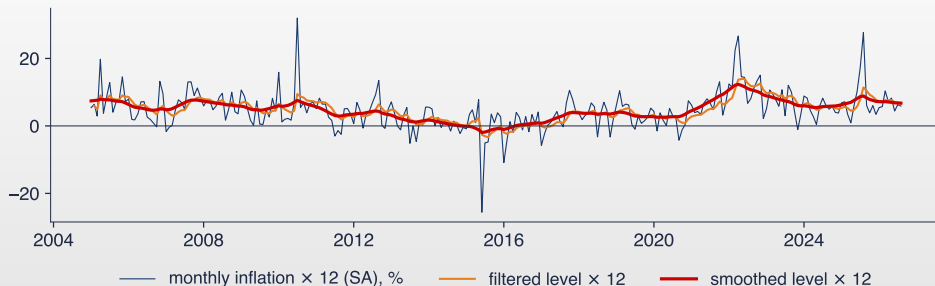
- ▣ The model of A5 with $\mu_1 = -0.5$, $\mu_2 = 1$, $\sigma = 1$; last quarter $\Pr(S_{t-1} = 1 \mid Y_{t-1}) = 0.2$; this quarter $y_t = -1$. Model: A5.
- ▣ 1. Compute the predicted probability $\Pr(S_t = 1 \mid Y_{t-1})$.
- ▣ 2. Compute the densities $f_1(y_t)$, $f_2(y_t)$ (standard Normal $\varphi(0.5) = 0.352$, $\varphi(2) = 0.054$) and the filtered probability.
- ▣ 3. Give the likelihood contribution and the predicted probability for next quarter.
- ▣ Report: four probabilities and one log-density.



B1: underlying Romanian inflation [Solved]

- **Question:** how fast does the underlying level of Romanian inflation move, and which exponential smoothing weight does the data choose?
- **Inputs:** monthly HICP inflation, seasonally adjusted, 2005–2026 ($T = 260$)
- **Tasks**
 1. Fit the local level model by maximum likelihood and report $\hat{\sigma}_\varepsilon^2$, $\hat{\sigma}_\eta^2$ and $\hat{q}$.
 2. Compute the steady-state gain $\bar{K}$ and compare it with the SES weight estimated directly.
 3. Plot the filtered and smoothed level ($\times 12$, in % per year) and report its peak and its last value.
 4. Interpretation: what does the size of $\hat{q}$ say about monthly inflation?
- **Report:** three variances, two weights, two levels, one sentence
- **Notebook:** section B1

B1: solution [Solved]



- $\hat{\sigma}_\varepsilon^2 = 0.142$, $\hat{\sigma}_\eta^2 = 0.0057$, $\hat{q} = 0.040$; $\bar{K} = 0.181$; SES by least squares: $\hat{\alpha} = 0.175$
- Smoothed level $\times 12$: peak 12.3% (April 2022), low -2.0% (June 2015), last 6.8% (August 2026, SE 1.9); the raw series $\times 12$ has SD 5.6
- Interpretation: q is small: most monthly movements are noise (tax changes, food and energy prices); each month moves the underlying level by about 0.181 of the surprise

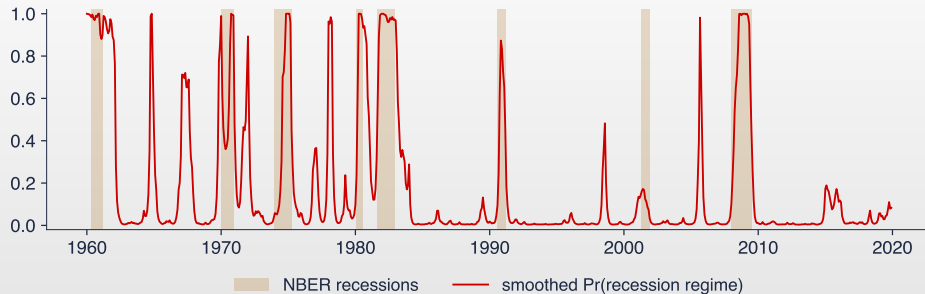
B2: the US output gap [Proposed]

- ▣ **Question:** how close are three statistical output gaps to the official CBO gap, and which one would you use in real time?
- ▣ **Inputs:** US real GDP and the CBO potential GDP, quarterly, 1949–2026; model: B1
- ▣ **Tasks**
 1. Compute the UC cycle (smooth trend plus AR(2)), the HP gap ($\lambda = 1600$) and the Hamilton filter ($h = 8$, four lags) of 100 log GDP.
 2. Compute the CBO gap $100(\text{GDP}/\text{GDPPOT} - 1)$ and the correlation of each measure with it, including the filtered UC cycle.
 3. Report the standard deviations and the latest values.
 4. Interpretation: which measure would you use to judge the economy today?
- ▣ **Report:** five correlations, four standard deviations, four latest values, one sentence
- ▣ **Notebook:** section B2

B3: recessions in US industrial production [Solved]

- ▣ **Question:** does a two-regime model of industrial production find the NBER recessions?
- ▣ **Inputs:** monthly growth of US industrial production, 1960–2019 (93 NBER recession months)
- ▣ **Tasks**
 1. Fit a two-regime model with switching mean and variance (`MarkovRegression`).
 2. Report the regime means and standard deviations, p_{11} , p_{22} and the expected durations.
 3. Classify each month by the smoothed probability (> 0.5) and by the filtered probability, and compare with the NBER months.
 4. Interpretation: does the recession regime match the NBER recessions?
- ▣ **Report:** six parameters, two durations, two concordance tables, one sentence
- ▣ **Notebook:** section B3

B3: solution [Solved]



- ▣ Recession regime: mean -0.16% , SD 1.30; expansion: mean 0.29% , SD 0.51; $\hat{\rho}_{11} = 0.874$, $\hat{\rho}_{22} = 0.970$; durations 8.0 and 33.2 months; ergodic share 19%
- ▣ Smoothed: 89% of months agree, 69% of NBER months found, 7% false alarms; filtered: 86%, 57%, 10%
- ▣ Interpretation: yes for the deep recessions; the regime is "low and volatile growth", so it also flags some slowdowns, and in real time it finds fewer recession months

B4: volatility regimes of the BET [Proposed]

- **Question:** does the Bucharest market have calm and turbulent regimes, and are they the same as those of the S&P 500?
- **Inputs:** weekly log returns of the BET and of the S&P 500, 2000–2026; model: B3
- **Tasks**
 1. Fit a two-regime model with switching mean and variance to BET returns; report the annualised volatilities and durations.
 2. List the three years with the largest share of turbulent weeks.
 3. Fit a GARCH(1,1) ([Chapter 5](#)) and correlate its volatility with the regime-implied volatility.
 4. Interpretation: are the turbulent periods of the BET those of the S&P 500?
- **Report:** two volatilities, two durations, three years, two correlations, one sentence
- **Notebook:** section B4

C1: regimes of Romanian inflation [Proposed]

- ▣ **Question:** how many inflation regimes has Romania had since 1996, and when did the low-inflation regime begin?
- ▣ **Inputs:** monthly HICP inflation, not seasonally adjusted, 1996–2026; inflation targeting since August 2005; model: B3
- ▣ **Tasks**
 1. Fit models with two and three regimes (switching mean and variance) and compare them by BIC.
 2. For three regimes, report the means, durations and the date when the low regime becomes permanent.
 3. Propose one more check (seasonality, a regime with an AR term, out-of-sample forecasts).
 4. Interpretation: did inflation targeting start the low-inflation regime?
- ▣ **Report:** six parameters, a date, a check and a project plan
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant about state space and regime models. The answer:
 - ▣ (a) For the Nile, $q = 0.097$, so the equivalent exponential smoothing weight is $\alpha = q = 0.097$.
 - ▣ (b) The Kalman gain is $K = \sigma^2_{\epsilon} / (P + \sigma^2_{\epsilon})$: the noisier the data, the more weight they get.
 - ▣ (c) With $p_{11} = 0.75$, recessions last $p_{11} / (1 - p_{11}) = 3$ quarters on average.
 - ▣ (d) To call a recession in real time, use `smoothed_marginal_probabilities` from `statsmodels`.
 - ▣ (e) The last value of the HP gap is a reliable estimate of today's output gap.
 - ▣ (f) In the local level model the forecasts of all horizons are equal; only their variance grows with the horizon.
- ▣ **Tasks**
 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (Part A, notebook section C2).
- ▣ **Report:** a list of six verdicts with one line of justification each

Key takeaways

- ▣ Kalman filter: $K_t = P_t / (P_t + \sigma_\varepsilon^2)$, $a_{t|t} = a_t + K_t v_t$; a missing value means no update
- ▣ Exponential smoothing is a local level filter in steady state: $\alpha = \bar{K}$, $q = \alpha^2 / (1 - \alpha)$
- ▣ Romanian monthly inflation: small q , a slowly moving underlying level
- ▣ Markov switching: durations $1 / (1 - p_{ii})$; filtered probabilities for real time, smoothed ones for history
- ▣ An AI answer is a draft: check the gain, the duration formula, filtered against smoothed

After the seminar

- Lecture 10 develops each topic of today
 - ▶ the state space form, the filter and the smoother, the likelihood, trend and cycle, Markov switching
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation regimes in Romania and in the region
- Reading: [Durbin and Koopman \(2012\)](#), Ch. 2; [Hamilton \(1989\)](#); [Hamilton \(2018\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

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